REAR chain: conflict with a following vehicle

Derivation runs downward, from the collision (lagging SPI) back to the earliest observable signals

LAGGING SPI: REAR collision

{ No. of times a REAR collision occurs per 100k manoeuvres } $< T_{REAR}$

Metric: contact between the following vehicle and the ego vehicle.

why

PROXIMATE CAUSE

The ego vehicle decelerated inside the manoeuvre, at turn entry or at a pedestrian stop, and the follower did not arrest its approach within the gap that remained. No third party is involved: the hazard is created by the ego vehicle's own deceleration.

reaction tier

NO REACTION-TIER INDICATOR ON THIS CHAIN

The ego vehicle has no last-resort mitigation directed rearward, so the reaction tier present on the ONC and PED chains is empty here.

decompose the proximate cause (kinematic precursor tier)

OR gate: either cause alone is sufficient

CAUSE A

The gap held by the follower was already too small to absorb a deceleration of the magnitude the manoeuvre requires.
Ego planning subsystem, through its speed profile.

CAUSE B

The follower responded late, continuing to close for an interval after braking began.
Follower's own driver or control stack.

SIGNAL A

Line-of-sight time to collision from the follower to the ego vehicle, sampled each frame.

SIGNAL B

Reaction delay of the following vehicle, measured from brake onset.

observability and independence check

LEADING SPI: REAR_TTC

{ No. of times REAR_TTC $< \theta_{REAR_TTC}$ per 100k manoeuvres } $< T_{REAR_TTC}$

Metric: minimum line-of-sight TTC, follower to ego vehicle.

REJECTED AT STEP 5

Not observable to the ego vehicle: the quantity belongs to the following driver and is available in simulation only because the delay is a sampled parameter.

ATTACH TO THE SAFETY CASE

REAR_TTC on G4.1 as leading evidence and the REAR collision as lagging evidence on G4, the following-conflict goal. The rejected branch corresponds to the undeveloped goal G4.2.

VALIDATE

Scored only on runs in which a follower is present, since the hazard cannot arise without one. The acceptance criterion is argued on the baseline-comparison principle, the following driver being a participant in the same traffic system.